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# Volatilities of different time resolutions – Analyzing the dynamics of market components

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**Abstract**

The diversity of agents in a heterogeneous market makes volatilities of different time resolutions behave differently. A lagged correlation study reveals that statistical volatility defined over a coarse time grid significantly predicts volatility defined over a fine grid. This empirical fact is not explained by conventional theories and models. We propose a new model class that takes into account squared price changes from time intervals of different size. This model is shown to reproduce the same empirical properties that have been found for FX intra-day data: long memory, fat-tailed distribution, and predictability of finely defined volatility by coarsely defined volatility. © 1997 Elsevier Science B.V.

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## 1. Introduction

A new idea in modeling financial markets is the hypothesis of a *heterogeneous market* where the market agents differ in their perceptions of the market, risk profiles, institutional constraints, degree of information, prior beliefs, and other

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characteristics such as geographical locations. In (Müller et al., 1993), we argue that many differences among market participants translate to a sensitivity to different time horizons. In the same paper, some empirical evidence for this hypothesis is collected from the foreign exchange (FX) market. The long memory of volatility as already found in Dacorogna et al. (1993) and Ding et al. (1993) is explained in terms of different market participants with different time horizons, from short-term FX dealers to long-term investors.

There is a growing number of papers dealing with different types of traders. Most of them are focused on aspects other than the time horizons of traders: informed traders/liquidity traders (Admati and Pfleiderer, 1988; Easley and O'Hara, 1992; Lyons, 1995), heterogeneous beliefs (Morris, 1994), herding of speculator groups on specific types of information (Froot et al., 1992). All of these recent papers reject or go a step beyond the classical view of rational agents that interpret information in the same way and have no reason to differ in their time horizons. In Froot et al. (1992), there is a discussion of this contradiction.

In our paper, we are focusing on the time horizon and the temporal resolution with which different traders are viewing and influencing the market [1]. We believe that the time horizon is one of the most important aspects in which trading behaviors differ and give some evidence for this. Aside from Müller et al. (1993), we find Bjorn (1994) sharing this point of view. Osler (1995) suggests that speculators do not specialize in a time horizon but rather adapt their horizon according to the changing relative attractiveness of short- and long-term speculation. In our words, this means that individual traders may sometimes leave a certain trader group and join another one with a different horizon.

Our basic idea is that different market agent types or *components* perceive, react to, and cause different types of *volatility*. Short-term traders evaluate the market at a higher frequency and have a shorter memory than long-term traders. Thus, we divide not only the market agents but also volatility into components. A similar idea is presented in Engle and Lee (1993) where two volatility components ('permanent' and 'transitory') are modeled without relating these to specific trader groups. The heterogeneous market hypothesis leads to the following statements on volatility:

- Volatilities (for example, mean absolute returns) measured with different time resolutions reflect the perceptions and actions of different market components.
- The interval size of the time grid on which volatility is measured is therefore an essential parameter. Volatility measured with high resolution contains information that is not covered by low resolution volatility and vice versa.

[1] There is also a methodological reason for this: the time horizon aspect can be investigated by studying the time series of prices whereas the study of other properties of trader groups often requires some less easily obtainable and quantifiable information.

● Studying the interrelations of volatilities of different time resolutions reveals the dynamics of market components.

An important method of interrelating different volatilities is lagged correlation. This method detects asymmetries in the information flow between components or even causal relations.

The goal of this paper is

● to present a new empirical fact: coarsely defined volatility predicts finely defined volatility significantly better than the other way around (asymmetry in the lagged correlation of volatilities of different time resolutions),

● to introduce a new generation process, HARCH, that can explain this fact and reflects the component structure of the market in a natural way,

● to identify the structure and relative presence of market components by estimating the HARCH parameters,

● to compare synthetic HARCH and GARCH time series with an FX time series and to show that only HARCH can explain the asymmetry in the lagged correlation of volatilities,

● in general: to increase the evidence in favor of the heterogeneous market hypothesis and to gain a deeper insight into the structure of such a market.

The study is done with intra-daily FX data which offers the great advantage of large samples and high significance. Some knowledge gained from Müller et al. (1990) and Dacorogna et al. (1993) is applied, in particular the deseasonalization of intra-daily volatility patterns with the help of the $\vartheta$-time scale presented in the latter paper. However, the essential finding of the paper does not depend on intra-day techniques such as changing time scales; it is also demonstrated in the case of daily data. From a market description point of view, the advantage of estimating the process on intra-day data is that the short-term components of the market are fully integrated and evaluated by the model.

In Section 2 of this paper, volatilities of different time resolutions are introduced and studied, leading to the detection of asymmetric lagged correlation. The HARCH process of Section 3 models this effect in a natural way. Its parameters are estimated in Section 4; they directly reflect the component structure of the heterogeneous market. A comparative study of HARCH and GARCH in Section 5 shows that only HARCH can model the lagged correlation effect and at the same time some other known facts such as the long memory of volatility. The conclusions are stated in Section 6.

## 2. Heterogeneous markets and lagged correlation of volatilities

### *2.1. Volatilities of different time resolutions*

The heterogeneous market hypothesis has been presented in Müller et al. (1993) and associated with fractal phenomena in the empirical behavior of FX markets. A

scaling law relating time horizon and size of price movements (volatility) has been identified in Müller et al. (1990). This relation is used here to explain why the perception of volatility differs for market agents with different time horizons.

Short-term traders are constantly watching the market; they re-evaluate the situation and execute transactions at a high frequency. Long-term traders may look at the market only once a day or less frequently. A quick price increase of 0.5% followed by a quick decrease of the same size, for example, is a major event for an FX intra-day trader but a non-event for central banks and long-term investors [2]. Long-term traders are interested only in large price movements and these normally happen only over long time intervals, see the scaling law of Müller et al. (1990). Therefore, long-term traders with open positions have no need to watch the market every minute [3]. In other words, they judge the market, its prices and also its volatility with a coarse time grid. A coarse time grid reflects the view of long-term trader and a fine time grid that of a short-term trader. A recent paper (Bjorn, 1994) follows similar arguments on market components.

The time grid in which real traders watch the market is not strictly regular, of course. In the following lagged correlation study, however, we measure volatilities over different but regularly spaced grids. These volatilities are defined as statistical volatilities: mean absolute price changes. We prefer mean absolute values to roots of mean squares here because they are statistically less dominated by extreme observations which are rather important in FX markets with their fat-tailed unconditional distribution functions. The convergence of the fourth moment — a requirement for many types of analysis such as the autocorrelation of squared returns — is doubtful for empirical returns. This is shown in Dacorogna et al. (1994), where the autocorrelation of absolute return is also shown to be much more stable under sample size changes than that of squared returns. The use of absolute returns is also found to be optimal in the autocorrelation study of Ding et al. (1993).

The statistical volatility based on absolute price changes has two essential timing parameters (Guillaume et al., 1994):

(1) The interval size of the time grid in which price changes are observed.

(2) The total size of the sample over which it is computed (the number of grid intervals considered).

### 2.2. Lagged correlation – The method

Analyzing the correlation between two time series, in our example fine and coarse volatility, is a standard tool used in empirical time series studies. The

[2] Small, short-term price moves may sometimes have a certain influence on the timing of long-term traders' transactions but never on their investment decisions as such.

[3] They have other means to limit the remaining risk of rarely occurring, fast and extreme price changes: stop-loss or options.

correlation coefficient is a straightforward, linear measure of the dependence of the two time series variables.

*Lagged* correlation is a more powerful tool to investigate the relation between two time series. The lagged correlation function considers the two series not only simultaneously (at lag 0) but also with a time shift. The correlation coefficient $\varrho_\tau$ of one time series and another one shifted by a positive or negative time lag $\tau$ is measured and plotted against the value of the lag.

Lagged correlation reveals causal relations and information flow structures in the sense of Granger causality. If two time series were generated on the basis of a synchronous information flow, they would have a symmetric lagged correlation function, $\varrho_\tau = \varrho_{-\tau}$; the symmetry would be violated only by insignificantly small, purely stochastic deviations. As soon as the deviations between $\varrho_\tau$ and $\varrho_{-\tau}$, become significant, there is asymmetry in the information flow and a causal relation that requires an explanation.

### 2.3. Empirical results: Asymmetry in the information flow

The logarithmic middle price quoted from an FX market as defined in Müller et al. (1990) and Guillaume et al. (1994) is the variable under investigation. In a first analysis, we consider a working-daily time series where weekends are omitted. Two variables are analyzed: 'fine volatility' and 'coarse volatility'. Fine volatility is the mean absolute working-daily price change averaged over 5 observations, so covering a full (working) week. Coarse volatility is the absolute price change over a full weekly interval. The essential results of this lagged correlation study are found also for other choices of analysis intervals, as to be shown below.

The correlation between fine volatility and coarse volatility is computed as a function of the lag. In the case of lag 0, the two corresponding time intervals observed for fine and coarse volatility are completely identical. In the case of lag 1 or $-1$, the two intervals do not overlap but follow each other immediately.

Fig. 1 shows the lagged correlation function for USD–DEM in a sample longer than 21 years. The correlation maximum is found at lag 0 — an expected result in this case of identical observation intervals. For the nonzero lags, we observe asymmetry: coarse volatility predicts fine volatility better than the other way around. The asymmetry is significant for the first two lags: the difference $\varrho_\tau - \varrho_{-\tau}$, represented by the thin curve in Fig. 1, is distinctly outside the confidence range of a Gaussian random walk.

This result can be explained in the sense of the heterogeneous market hypothesis (see Section 2.1). For short-term traders, the level of coarse volatility matters because it determines the expected size of trends and thus the scope of trading opportunities. Short-term traders react to clusters of coarse volatility by changing their trading behavior and so causing clusters of fine volatility. On the other hand, the level of fine volatility does not affect the trading strategies of long-term traders (who are often considering the 'fundamentals' of the market).

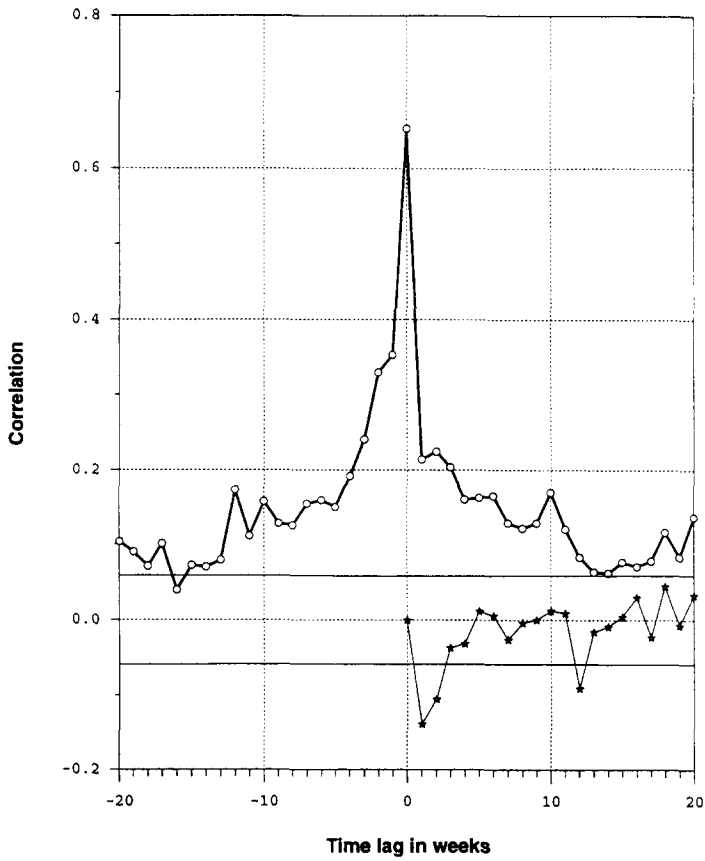



Fig. 1. Asymmetric lagged correlation of fine and coarse volatilities of a working-daily USD–DEM series. The fine volatility is defined as the mean absolute working-daily price change in a week; the coarse volatility is the absolute price change over a full week. The thin curve indicates the asymmetry: the difference between correlations at positive and corresponding negative lags. Sampling period: 21 years and 8 months, from 6 Jun 1973 to 1 Feb 1995. The confidence limits represent the 95% confidence interval of a Gaussian random walk.

Similar behavior of the lagged correlation is observed for other FX rates such as USD–JPY, GBP–USD, cross rates such as DEM-JPY, and gold (XAU-USD). In Table 1 we report the difference $\varrho_1 - \varrho_{-1}$ and $\varrho_2 - \varrho_{-2}$ for a set of these time series. The numbers are similar across the different rates (and also all the investigated minor FX rates not shown here). The first lag difference is around $-0.13$ and the second lag difference around $-0.07$.

The results of this study of daily data can be strongly confirmed by analyzing high-frequency, intra-day data. Every intra-day study requires an appropriate treatment of the strong intra-daily seasonality of volatility. Here, we use the

Table 1
Differences between lagged correlations of fine and coarse volatilities, for different FX rates and gold

| Differences | USD–DEM | USD–JPY | GBP–USD | CHF–USD | DEM–JPY | XAU–USD |
|---|---|---|---|---|---|---|
| $\varrho_1 - \varrho_{-1}$ | −0.138 | −0.127 | −0.130 | −0.131 | −0.129 | −0.122 |
| $\varrho_2 - \varrho_{-2}$ | −0.105 | −0.047 | −0.055 | −0.076 | −0.074 | −0.072 |

The sampling of the time series and the type of analysis are the same as in the case of Fig. 1; the sampling period is from 6 Jun 1973 (8 Aug 1980 for gold) to 1 Feb 1995, and the lags are measured in weeks. The consistent, negative values indicate the predictability of finely defined volatility from coarse volatility.

predefined business time scale $\vartheta$: the volatility is deseasonalized by expanding time at daily volatility peaks and contracting it during inactive periods (Dacorogna et al., 1993). A time series with regular intervals in $\vartheta$-time is constructed by selecting the last quote before each point of a regular $\vartheta$-grid [4].

As a basic time interval in $\vartheta$-time, we choose 30 min. This means only some 7 min of physical time during the daily volatility peak in the European afternoon and American morning. In fact, a much higher frequency of the series should be avoided: price changes observed over 5 min or less can be overly biased by the buying and selling intentions of the quoting institutions (Guillaume et al., 1994). Fine volatility is now the mean absolute half-hourly price change averaged over 6 observations, so covering a 3 h time interval. Coarse volatility is the absolute price change over a full 3 h interval. All these time intervals are meant in $\vartheta$-time. An interval of 3 $\vartheta$-hours is clearly smaller than the working day of an FX dealer, it often covers a time span with quite homogeneous market conditions.

Fig. 2 shows the lagged correlation function for USD–DEM in a sample longer than 8 years. Although this sample is shorter than that of the daily data, the number of observations is higher thanks to the much shorter basic interval of the intra-daily series. The higher number of observations leads to a narrower confidence interval and thus less noise in the lagged correlation curve. The message is the same as in the daily data case but reinforced by even higher significance: coarse volatility predicts fine volatility significantly better than the other way around. We conclude that this is rather independent of the chosen interval sizes.

Similar intra-daily behavior of the lagged correlation is reported for other FX rates and gold, see Table 2. The numbers are very similar across the different rates; the first lag difference is around −0.11 and the second lag difference around −0.06, quite close to the corresponding values of Table 1.

In Fig. 2, we can also find a weak, rather wide local maximum around lag −11 , corresponding to minus 33 h in $\vartheta$-time. This means a lag of about one working day (because a working day is 1/5 rather than 1/7 of a business week). The

[4] The choice of the first point of the $\vartheta$-grid determines the phase shift of the grid. We find a low sensitivity of the results against phase shifts with only stochastic, insignificant changes.

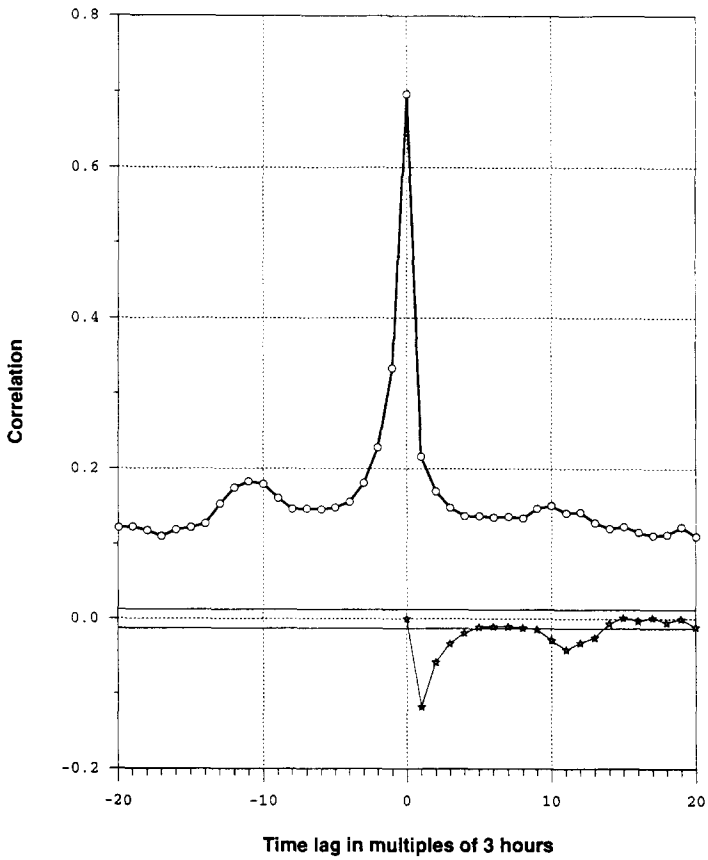



Fig. 2. Asymmetric lagged correlation of fine and coarse volatilities of a half-hourly USD–DEM series in $\vartheta$-time. The fine volatility is defined as the mean absolute half-hourly price change within 3 h ($\vartheta$-time); the coarse volatility is the absolute price change over a whole 3 h interval ($\vartheta$-time). The thin curve indicates the asymmetry: the difference between correlations at positive and corresponding negative lags. Sampling period: 8 years, from 1 Jan 1987 00.00 to 1 Jan 1995 00.00 (GMT). The confidence limits represent the 95% confidence interval of a Gaussian random walk.

Table 2
Differences between lagged correlations of fine and coarse volatilities, for different FX rates and gold

| Differences | USD–DEM | USD–JPY | GBP–USD | CBF–USD | DEM–JPY | XAU–USD |
|---|---|---|---|---|---|---|
| $\varrho_1 - \varrho_{-1}$ | −0.117 | −0.136 | −0.113 | −0.093 | −0.100 | −0.108 |
| $\varrho_2 - \varrho_{-2}$ | −0.058 | −0.057 | −0.059 | −0.056 | −0.055 | −0.068 |

The sampling of the time series and the type of analysis are the same as in the case of Fig. 2; the sampling period is from 1 Jan 1987 00.00 to 1 Jan 1995 00.00 (GMT). The consistent, negative values indicate the predictability of finely defined volatility from coarse volatility.

difference $\varrho_\tau - \varrho_{-\tau}$, also has a significant (negative) peak around lag 11. This effect has been identified already in Dacorogna et al. (1993) and discussed in Müller et al. (1993). Following Engle et al. (1990), we call it a 'heat wave' effect: traders have a better memory of the events approximately one working day ago (when they were active) than a broken number of working days ago (when other traders on different continents, with different time zones were active).

The peak around lag $-11$ might be superficially explained by a residual seasonality that the $\vartheta$-scale is unable to capture. However, the $\vartheta$-scale is well able to treat ordinary seasonality as indicated by the lack of an analogous peak around the positive lag 11. The heat wave effect is more than just seasonality; it cannot by eliminated by a simple time scale transformation. This leads to a new idea for future work: volatility modeling should consider not only volatilities of different time resolutions but also volatilities with the selective memory of individual geographical markets and their time zones.

## 3. HARCH, a process to model heterogeneous volatilities

### *3.1. The basic idea*

The asymmetric behavior of fine and coarse volatilities as found in Section 2.3 cannot be explained by known processes from the ARCH family. All these processes have a variance equation based on returns that are observed exclusively over intervals of the same size. The failure of GARCH to model the asymmetry is demonstrated below.

The new HARCH process proposed in this paper has a variance equation based on price changes over intervals of *different sizes*. The empirical behavior of lagged correlation can be reproduced well with this new process. At the same time, HARCH is able to reproduce the long memory of volatility [5] as found in Dacorogna et al. (1993) and Ding et al. (1993). Moreover, the terms of the conditional variance of HARCH reflect the component structure of the market in a natural way.

As most processes from the ARCH family [6], HARCH is based on squared price changes with their good analytical tractability. Whereas the existence problem of the fourth moment of the return distribution forced us to define volatility in terms of absolute returns in the correlation analysis of Section 2, there is no such constraint for the volatility equation of the HARCH process.

[5] The new FIGARCH process (Baillie et al., 1996) has been designed to model the long memory but cannot be expected to reproduce the lagged correlation as it is still based on returns observed over intervals of constant size.

[6] Except a process class proposed by Ding et al. (1993) which models volatility in terms of different powers of absolute returns.

### 3.2. Definition

The returns $r_t$ of a HARCH($n$) process are defined with the help of the random variable $\varepsilon_t$ which is i.i.d. and follows a distribution function with zero expectation and unit variance [7]:

$$r_t = \sigma_t \varepsilon_t, \qquad \sigma_t^2 = c_0 + \sum_{j=1}^{n} c_j \left( \sum_{i=1}^{j} r_{t-i} \right)^2 \tag{1}$$

where

$$c_0 > 0, \quad c_n > 0, \quad c_j \geq 0 \qquad \text{for } j = 1, \ldots, n-1 \tag{2}$$

The equation for the variance $\sigma_t^2$ is a linear combination of the squares of *aggregated* returns. Aggregated price changes may extend over some long intervals from a time point in the distant past up to time $t-1$. The heterogeneous set of relevant interval sizes leads to the process name HARCH for 'heterogeneous interval, autoregressive, conditional heteroskedasticity'. The first 'H' may also stand for the heterogeneous market if we follow that hypothesis as proposed in Section 2.

The HARCH process belongs to the wide ARCH family but differs from all other ARCH-type processes in the unique property of considering the volatilities of price changes measured over different interval sizes. The Quadratic ARCH (QARCH) process (Sentana, 1991) is an exception: although it was not developed for treating different interval sizes, it can be regarded as a generalized form of HARCH as explained in Section 3.5.

The coefficients $c_1 \ldots c_n$ should not be regarded as free parameters of the model. The heterogeneous market approach leads to a low number of free model parameters which determine a much higher number $n$ of coefficients modeling the long memory of volatility.

The explicit formulation of HARCH(2) may help to illustrate the special properties of the HARCH process [8]. The variance equation of HARCH(2) can be written in two forms:

$$\begin{aligned} \sigma_t^2 &= c_0 + c_1 r_{t-1}^2 + c_2 (r_{t-1} + r_{t-2})^2 = c_0 + (c_1 + c_2) r_{t-1}^2 + c_2 r_{t-2}^2 \\ &\quad + 2 c_2 r_{t-1} r_{t-2} \end{aligned} \tag{3}$$

The second form of this HARCH(2) process can be identified as an ordinary ARCH(2) process — except for its last term which contains the mixed product $r_{t-1} r_{t-2}$. In other ARCH-type processes, the absolute values of returns matter; in HARCH, also their *signs* matter: two subsequent price changes of the same size

[7] In this paper, we take a normal distribution and alternatively test Student-$t$ distributions.

[8] Whereas HARCH(1) is identical to ARCH(1).

and in the same direction will cause a higher contribution to the variance process than two subsequent price changes that cancel out each other.

### 3.3. Stationarity and the existence of moments

The variance, the unconditional expectation of squared returns, can be derived from Eq. (1):

$$E(r_t^2) = E(\sigma_t^2) = c_0 + \sum_{j=1}^{n} c_j \left[ \sum_{i=1}^{j} E(r_{t-i}^2) \right] \tag{4}$$

There is no term with cross products here as their expectation is zero.

A necessary condition of stationarity is constant unconditional variance:

$$E(r_t^2) = E(r_{t-i}^2), \quad i \geq 1 \tag{5}$$

Inserting this in Eq. (4), we obtain the variance

$$E(r_t^2) = \frac{c_0}{1 - \sum_{j=1}^{n} jc_j} \tag{6}$$

which must be finite and positive:

$$\sum_{j=1}^{n} jc_j < 1 \tag{7}$$

This necessary stationarity condition is also a sufficient condition for both the stationarity of the process and the existence of the variance, the second moment. Proving this is not trivial, we do not follow here the path chosen by Engle (1982) and Bollerslev (1986) because the mixed products such as $r_{t-1}r_{t-2}$ make the matrix formulation of the problem difficult. The HARCH process can be seen as a Markov chain. Meyn and Tweedie (1993) have obtained some results for the ergodicity and recurrence of Markov chains that can be used for proving the stationarity and the moment condition. The complete proof is given by Dacorogna et al. (1995).

The conditions for the existence and constant unconditional expectation of higher moments can be obtained through steps analogous to Eqs. (4)–(7), but the computation becomes increasingly tedious for higher moments and larger $n$ values. The expectation of the $2m$th moment is

$$E(r_t^{2m}) = E(\sigma_t^{2m}) E(\varepsilon_t^{2m}) \tag{8}$$

The variance equation (Eq. (1)) is inserted in $E(\sigma_t^{2m})$ and all the terms explicitly computed. Some products of powers of returns have non-zero expectations, leading to an equation system for these expectations and $E(\sigma_t^{2m})$. The equation system has the dimension $m$ for $n = 2$ and higher for larger $n$ values.

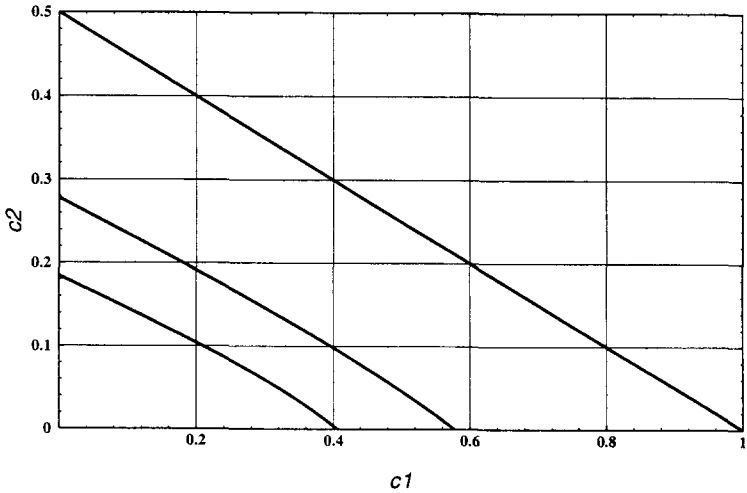



Fig. 3. Moment conditions for the coefficients $c_1$ and $c_2$ of a HARCH(2) process with a normally distributed $\varepsilon_t$. The straight line on the right represents the boundary for the stationarity and the existence of the second moment. The curves in the middle and on the left represent the boundaries for the existence of the 4th and 6th moments.

In the relatively simple case of the fourth moment ($m = 2$) of HARCH(2) ($n = 2$), the expectation $E(r_t^2 r_{t-1}^2)$ has to be computed and solved together with the equation for $E(r_t^4)$. In the standard case of $\varepsilon_t$ following a normal $N(0, 1)$ distribution ($E(\varepsilon_t^4) = 3$), the following necessary condition is obtained to keep the fourth moment finite:

$$3\left[c_2^2 + (c_1 + c_2)^2\right] + c_2\left[1 + 3\left(c_1^2 + 6c_1c_2 + 4c_2^2\right)\right] < 1 \tag{9}$$

The sufficiency of this necessary fourth moment condition is also proven in Dacorogna et al. (1995).

In Fig. 3, the second and fourth moment conditions according to Eqs. (7) and (9), plus the sixth moment condition following an analogously derived equation, are plotted for a HARCH(2) process. Processes with a finite second and a diverging fourth moment exist in a large part of the $C_1 - C_2$-plane. This is remarkable because half-hourly FX returns have an empirical distribution with a tail index between 2 and 4 (Dacorogna et al., 1994).

### 3.4. HARCH and market components – A parsimonious parametrization

The memory of the volatility autocorrelation is known to be long (Dacorogna et al., 1993). Therefore, we need a high order of HARCH, a large $n$, to model the behavior of empirical time series. This implies a high number of coefficients $c_j$ which should not be free parameters of HARCH. We need a parsimonious

parametrization. In the case of ARCH, some high-order processes can be formulated as low-order GARCH processes (Bollerslev, 1986), but no analogous method is at hand to reduce the number of HARCH parameters.

A possible approach could be to introduce a parametric relation between the coefficients, for example $c_j = k_j^{\mu}$. This hyperbolic equation with only two free parameters, $k$ and $\mu$, has been tried quite successfully, but it misses the chance of revealing the component structure of the market. The coefficients $c_j$ reflect the relative impact of different market components with different relevant time intervals. Therefore, we select $m$ market components corresponding to $m$ free parameters, each associated to some coefficients $c_j$ in a limited range of $j$. The $j$ ranges are separated by powers of a natural number $p$, so the typical time interval size of a component differs from that of the neighbor component by a factor $p$. All $c_j$ values within one component are assumed to be the same:

$$c_j = C_i = C_{i(j)},$$

$$i(j) = \max\left(k | k \in \mathbb{N} \wedge k < \frac{\log j}{\log p} + 2\right), \quad j = 1, \ldots, p^{m-1} \tag{10}$$

Only $m$ different coefficients $C_i$ have to be estimated to determine the whole set of $n = p^{m-1}$ coefficients $c_j$.

Table 3 presents such a component scheme for a time series in $\vartheta$-time with a basic grid of 30 min, $p = 4$, and 7 components ($m = 7$). An interval of 30 min in $\vartheta$-time means only some 7 min during the daily volatility peaks around 14.00 GMT, some 80 min during the Far Eastern lunch break, and even more during weekends and holidays with their very low volatility. Table 3 shows the minimum relevant time intervals of a component rather than the total size of the volatility memory. In fact, the memory of the volatility can greatly exceed the indicated interval. The medium-term traders of component 5, for example, are not interested in the volatility of hourly price changes, are most interested in volatilities observed over $1$–$3\frac{1}{2}$ days, and are also interested in volatilities observed over longer intervals.

The choice of the number of components, $m = 7$, and the factor between the typical time resolutions of the components, $p = 4$, is reasonable but somewhat arbitrary. The essential results of this paper do not strongly depend on this choice and can be found also with other $m$ and $p$ values. The model should cover the variety of relevant time resolutions of the market [9]. A too small choice of $m$ misses the chance of revealing the component structure; a too large $m$ (with a small $p$) leads to too many parameters to be estimated and an unrealistically narrow definition of market components.

[9] The choice of Table 3 is $m = 7$. An even higher value, $m = 8$, has also been tested, leading to a low, insignificant impact of the eighth component and a rejection in a likelihood ratio test. We conclude that the seventh component is the last relevant one on the long-term side.

Table 3
A HARCH model with 7 market components, each with a range of indices $j$

| $i$ | Range of $j$ | Approximate range of time intervals, in $\vartheta$-time | Shortest interval (at daily or weekly volatility peaks), in physical time | Longest interval (but avoiding weekends and holidays), in physical time | Tentative description of the component |
|---|---|---|---|---|---|
| 1 | 1 | 30 min | 7 min | 80 min | short-term, intra-day dealers (arbitrage opportunities), market makers |
| 2 | 2–4 | 1–2 h | 16 min | $3\frac{1}{2}$ h | intra-day dealers with few transactions per day |
| 3 | 5–16 | $2\frac{1}{2}$–8 h | 50 min | 9 h | dealers with overnight positions and occasional intra-day transactions |
| 4 | 17–64 | $8\frac{1}{2}$–32 h | 4 h | 1 day | few traders concerned (time intervals often beyond local business hours but less than a business day) |
| 5 | 65–256 | $1\frac{1}{2}$–5 days | 1 day | $3\frac{1}{2}$ days | medium-term traders, no intra-day trading |
| 6 | 257–1024 | $5\frac{1}{2}$-21 days | $3\frac{1}{2}$ days | 21 days (weekends always contained) | long-term traders |
| 7 | 1025–4096 | 3–12 weeks | 3 weeks | 12 weeks (weekends always contained) | long-term investors, central banks |

All coefficients $c_j$ of a component are identical, so only 7 parameters have to be estimated. The time intervals are the relevant intervals for the volatility perception of the components (not the total duration of their memory). These basic intervals are given in $\vartheta$-time and also in physical time: two columns show the minimum and maximum size of the interval that can occur for a component, depending also on the daytime. The component descriptions, although not scientifically rigid, may contribute to a better understanding of the model.

The coefficients $C_i$ associated to the components have to be estimated as discussed below. A more suitable quantity for the intuitive understanding is the impact of different components on the volatility process. The expected variance formula, Eq. (6), strongly suggests to define the impact of the $j$th coefficient as $jc_j$. The impact $I_i$ of the $i$th component is defined as the sum of the impacts of all its coefficients $c_j$. By inserting the coefficient definitions of Eq. (10), we obtain

$$I_1 = c_1 = C_1,$$

$$I_i = \sum_{j=p^{i-2}+1}^{p^{i-1}} jc_j = (p-1)p^{i-2}\frac{p^{i-1}+p^{i-2}+1}{2}C_i \quad \text{for } i > 1 \tag{11}$$

The impact of the long-term components may be considerable even when the coefficients appear to be small. The impact of the fifth component, for example, is $I_5 = 30816C_5$, where $p$ is assumed to be 4 as in Table 3.

The stationarity condition, Eq. (7), can now be formulated in terms of the impacts. Their sum is smaller than 1:

$$\sum_{i=1}^{m} I_i < 1 \tag{12}$$

### 3.5. Generalization of the process equation

In Eq. (1), all the price changes considered by the variance equation are observed over 'recent' intervals ending at time $t-1$. This strong limitation will be justified by its empirical success, but we can also formulate a more general process equation with observation intervals ending in the past, before $t-1$:

$$r_t = \sigma_t \varepsilon_t, \qquad \sigma_t^2 = c_0 + \sum_{j=1}^{n}\sum_{k=1}^{j} c_{jk}\left(\sum_{i=k}^{j} r_{t-i}\right)^2 + \sum_{i=1}^{q} b_i \sigma_{t-i}^2, \tag{13}$$

where

$$c_0 > 0;$$

$$c_{jk} \geq 0 \quad \text{for } j = 1, \ldots, n, \quad k = 1, \ldots, j; \qquad b_i \geq 0 \quad \text{for } i = 1, \ldots, q \tag{14}$$

This generalized process equation considers all price changes between any pair of two time points in the period between $t-n$ and $t-1$. It covers the case of HARCH (all $c_{jk} = 0$ except some $c_{j1}$) as well as that of ARCH and GARCH (all $c_{jk} = 0$ except some $c_{jj}$). The last term of the variance equation is a 'GARCH term' which may contribute to a parsimonious model formulation. Such a GARCH term may partially model the fading volatility memory of several market components together, but therefore miss the chance of clearly indicating the actual

component structure. The main idea of HARCH, taking intervals of different sizes, may also be combined with other ideas from the recent literature about GARCH variations.

HARCH can also be regarded as a special case of the Quadratic ARCH (QARCH) model suggested by Sentana (1991). The results obtained for QARCH also apply to HARCH. However, QARCH has been developed in a very different context. Sentana (1991) gives neither a concept of volatilities observed over long intervals nor the stationarity and moment conditions as in Section 3.3.

In this paper, the simple form of Eq. (1) is preferred. The simple HARCH is successful enough in our empirical studies, so no motivation is left for taking a more complicated model such as Eq. (13) with its many more parameters.

## 4. Parameter estimation of HARCH – Revealing market components

A total number of 8 parameters has to be estimated: $c_0$ and $C_1 \ldots C_7$ (the coefficients of the components of Table 3). The random variable $\varepsilon_t$ of Eq. (1) is assumed to be normally distributed. In an alternative approach, we have also estimated a ninth parameter, the degree of freedom $\nu$ of the Student-$t$ distribution of the random variable $\varepsilon_t$, following a suggestion by Bollerslev (1987). The parameters are estimated to fit a half-hourly time series of USD–DEM, regularly spaced in $\vartheta$-time over 7 years, from 5 Jan 1987 to 2 Jan 1994.

The maximum of the log-likelihood function was searched with the BHHH algorithm (Berndt et al., 1974), where the initial guess of parameter values came from a prior search with a genetic algorithm. The convergence of the BHHH algorithm was good for all investigated cases, various FX rates and $\varepsilon_t$ distributed normally or Student-$t$ — much better than for GARCH estimation on the same FX data. This good convergence can be attributed to the fact that the various terms, which contribute to the conditional variance, are distinctly different. Each term represents a different market component, so the Hessian matrix of the optimization in the parameter space is far from singular.

Table 4 shows the resulting parameter estimates in the case of USD–DEM and a normal distribution [10] of $\varepsilon_t$. All parameters are highly significant, so all components of the model significantly contribute to the HARCH variance equation.

Instead of the coefficients $C_i$, the impacts $I_i$ are estimated (but the $C_i$ can be computed through Eq. (11)). Unlike the coefficients, the impacts can be directly compared. Their sum is 0.8176. The stationarity condition, Eq. (12) and thus Eq.

[10] In this paper we present the normal distribution case because HARCH is to be compared to GARCH, and the GARCH estimation with a Student-$t$ distribution on high-frequency data has bad convergence properties.

Table 4
Coefficients and impacts of a HARCH process, fitting a half-hourly USD–DEM series which is equally spaced in $\vartheta$-time over 7 years

| Coefficient | Estimate | Standard error | $t$-statistics |
|---|---|---|---|
| $c_0$ | $1.485 \times 10^{-7}$ | $0.092 \times 10^{-7}$ | 16.14 |
| $I_1$ | 0.1293 | 0.0128 | 10.13 |
| $I_2$ | 0.1899 | 0.0170 | 11.14 |
| $I_3$ | 0.1477 | 0.0146 | 10.08 |
| $I_4$ | 0.0592 | 0.0117 | 5.08 |
| $I_5$ | 0.1011 | 0.0131 | 7.75 |
| $I_6$ | 0.1102 | 0.0119 | 9.28 |
| $I_7$ | 0.0802 | 0.0094 | 8.56 |

Instead of the coefficients $C_i$, the impacts $I_i$ are given. These provide a direct measure of the impacts of the market components on the HARCH variance. The market components are those defined in Table 3. The coefficients $C_i$ or $c_j$ can be computed from $I_i$ through Eq. (11). The distribution of the random variable $\varepsilon_t$ is normal.

(7), is well fulfilled. A comfortable distance from the stationarity limit is typical for HARCH, whereas the coefficients of GARCH processes tend to be very close to the stationarity limit.

The significance of the model is further confirmed by likelihood ratio (LR) tests, following Hamilton (1994, p. 144). Thanks to the extremely large intra-day sample over 7 years, we can reject the null hypothesis of $I_1 = \ldots = I_7 = 0$ (a Gaussian random walk) with overwhelming significance. The more relevant null hypothesis of a HARCH with only 6 components ($I_7 = 0$) is also strongly rejected: the double log-likelihood difference is 597, and the probability of a $\chi^2(1)$ variable exceeding this high value is almost zero. The rejection of GARCH(1, 1) is even stronger [11]. On the other hand, a HARCH with 7 components cannot be rejected as the null hypothesis of a HARCH with 8 components ($m = 8$). The choice of 7 market components is confirmed.

The impacts $I_i$ are also plotted in a histogram, Fig. 4, where it is also possible to compare the results for different FX rates. The impact of the fourth component is the weakest among all impacts; this is found as a result of HARCH estimation for not only USD–DEM but also many other rates and also for other sampling periods [12]. In Table 3, we find that this fourth component considers price changes over intervals between 4 and 24 h (of physical time). This is the range of the least

[11] GARCH(1,1) is not a restricted version of HARCH, so the relevance of an LR test directly comparing these two models is debatable.

[12] We split the total 7 year sample into two parts of $3\frac{1}{2}$ years. The coefficients fitted on both of these subsamples are quite stable.

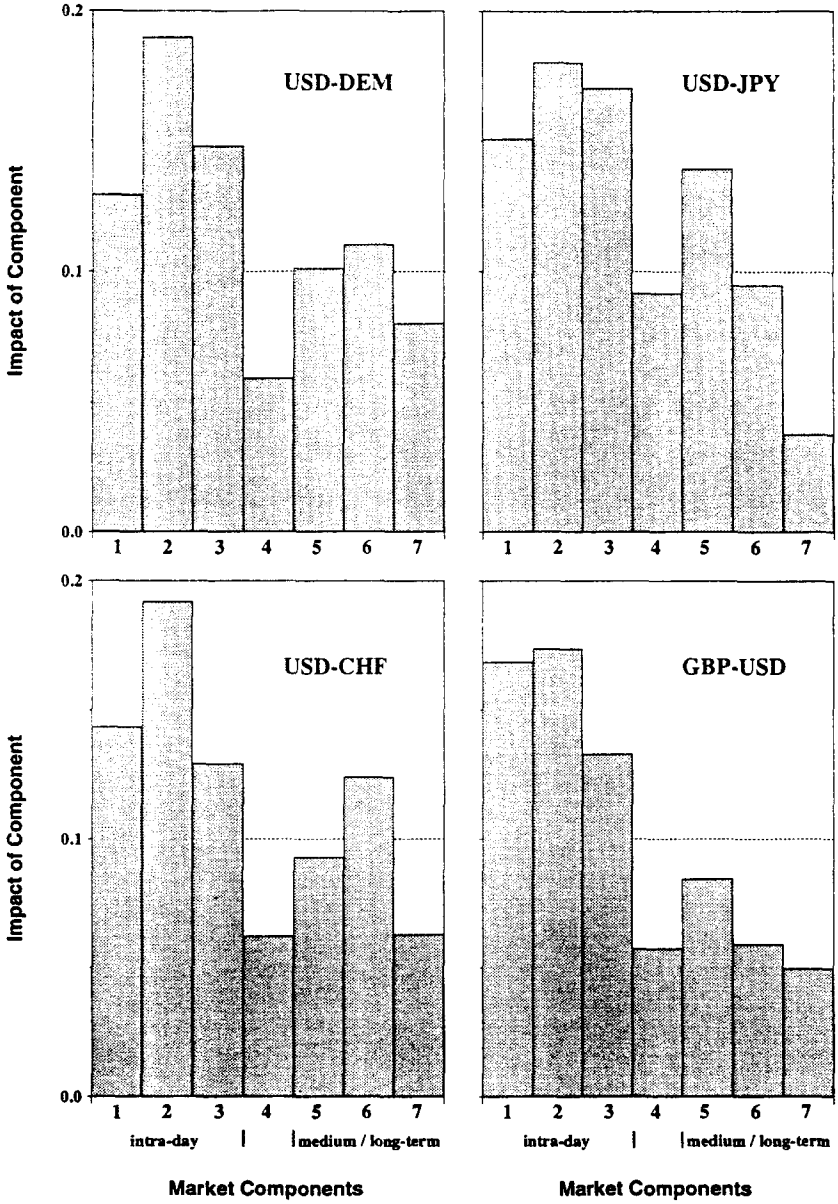



Fig. 4. Impacts of market components of HARCH processes with components as defined in Table 3. Each HARCH model has been made for a particular FX rate by fitting a half-hourly time series equally spaced in $\vartheta$-time over 7 years. The differences between the impacts, in particular the low values of the fourth component, are highly significant (see the error values of Table 4). The values for USD–DEM are those presented in Table 4; they are not fundamentally different from those of other FX rates.

interesting time intervals: intra-day traders look at shorter intervals, medium-term traders at daily or longer intervals. Few traders are interested in 12 h price changes which always compare time points at which traders on different sides of the globe are active. The empirically found weakness of the fourth component makes economic sense. Moreover, it leads to the idea of introducing a new type of market components: components that explicitly model the volatility perceptions of geographical markets such as America, Europe, or East Asia, with a low weight of the night periods of these markets, as already suggested in Müller et al. (1993). We have not yet tried to introduce this.

The representation of the HARCH parametrization in terms of market components, as given in Fig. 4, sheds a new light on many interesting features captured by our model. These features are known to many participants in the FX market. First, the short-term components have, in all cases, the largest impacts. These short-term components model essentially the intra-day dealers and the market makers who are known to dominate the market. Second, the similarity in the impacts of the USD–CHF and USD–DEM also makes good sense as it is well known that the Swiss National Bank policy is to tightly bind the USD–CHF to the USD–DEM and that both economies are also very close. The relative weakness of the longer term components for the GBP–USD is another relevant information that can be gathered from this parametrization and has been confirmed to us by market participants. Since the 1992 crisis, the long-term investors have been reluctant to invest in this market and have more concentrated on the cross rate GBP–DEM. The relative impact of the 5th and the 6th components are in the same order for USD–CHF and USD–DEM but inverted in the case of both USD–JPY and GBP–USD.

## 5. HARCH, GARCH, and real FX data – A comparative study

For this Monte Carlo simulation study, we have generated synthetic time series from the HARCH process [13] estimated in Section 4, see Table 4, and from a GARCH process which has been estimated on the same USD–DEM time series. The two resulting synthetic, half-hourly time series have an equivalent sample size

[13] The HARCH process with its long memory of 'aggregated' volatility is extremely sensitive against the serial correlation of $\varepsilon_t$ caused by a bad random number generator. Even some well-established random generators show some weak serial correlation, leading to HARCH series with biased, too high autocorrelations of absolute returns. This correlation can be broken only by randomly shuffling the insufficient random numbers in a huge buffer, leading to lower, unbiased autocorrelations of absolute returns. The autocorrelation of absolute returns of synthetic HARCH processes might in fact be used as a powerful test for detecting remaining serial correlation in random number generators.

of 7 years. This corresponds to 20455 intervals of 3 h and thus 20455 observations of coarse and fine volatility.

Since this paper is focused on the asymmetric lagged correlation of different volatilities, the same lagged correlation analysis as in Section 2.3 has been done for both synthetic series. Fig. 5 shows the lagged correlation function of fine and coarse volatility for the HARCH series. The similarity to the empirical lagged correlation for USD–DEM, Fig. 2, is striking. The asymmetry in the lagged correlation is highly significant for the first few lags. Except for the second weak peak around the lag $-11$ present in Fig. 2, which is attributed to geographical

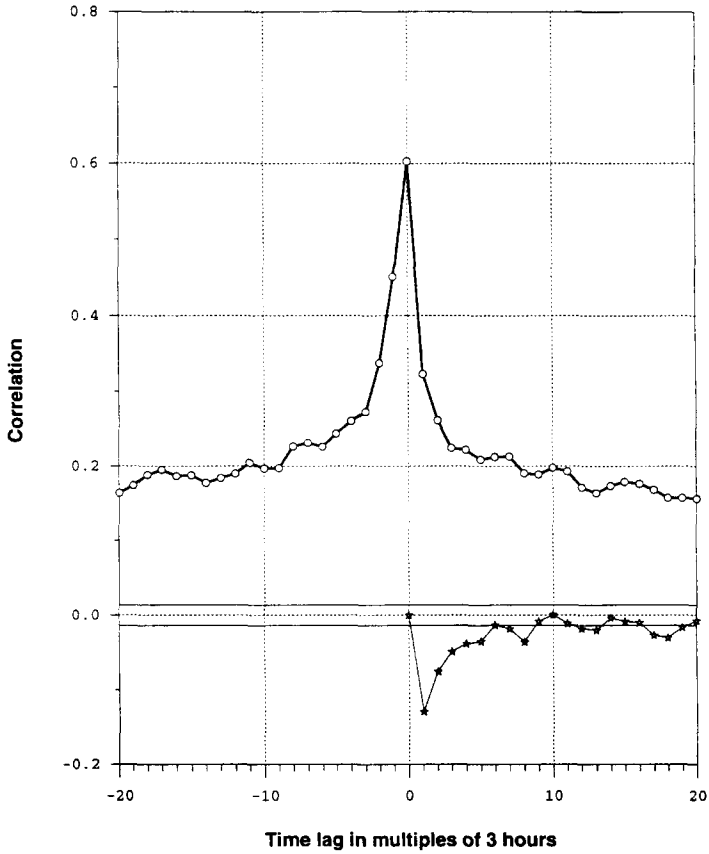



Fig. 5. Asymmetric lagged correlation of fine and coarse volatilities of a synthetic HARCH time series, fitted to a USD–DEM series on a half-hourly time grid in $\vartheta$-time. The fine volatility is defined as the mean absolute half-hourly price change in an interval of 3 h; the coarse volatility is the absolute price change over the whole 3 h interval. The thin curve indicates the asymmetry: the difference between correlations at positive and corresponding negative lags. Sampling period: 7 years. The confidence limits represent the 95% confidence interval of a Gaussian random walk.

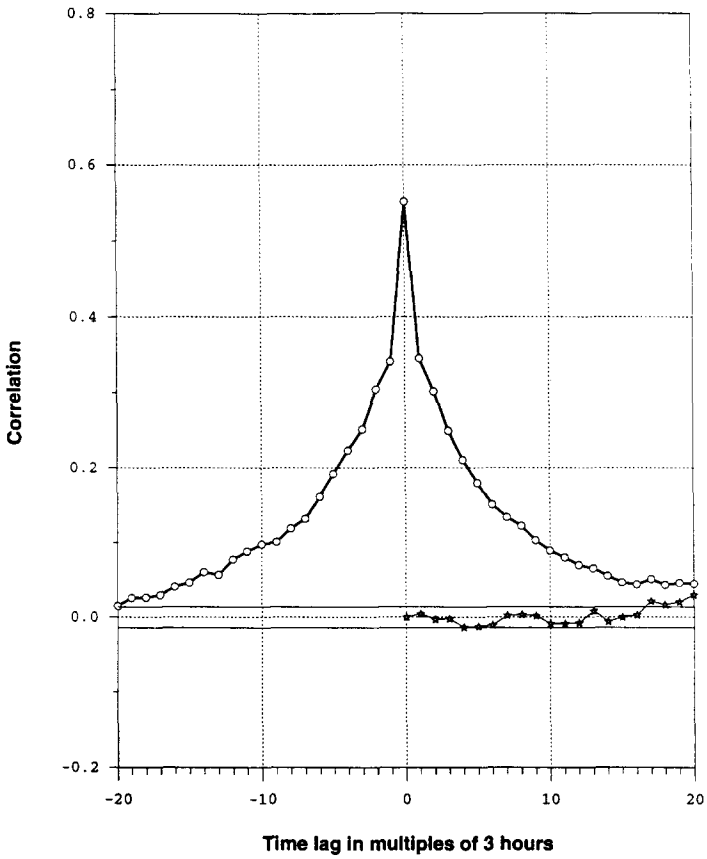



Fig. 6. Symmetric lagged correlation of fine and coarse volatilities of a synthetic GARCH time series, fitted to a USD-DEM series on a half-hourly time grid in $\vartheta$-time. The fine volatility is defined as the mean absolute half-hourly price change in an interval of 3 h; the coarse volatility is the absolute price change over the whole 3 h interval. The thin curve indicates the asymmetry: the difference between correlations at positive and corresponding negative lags (which is insignificant here). Sampling period: 7 years. The confidence limits represent the 95% confidence interval of a Gaussian random walk.

effects, the main effect is well reproduced. Again, we observe that coarse volatility predicts fine volatility better than the other way around.

The HARCH results are compared with those from a GARCH(1, 1) process [14].

[14] We also tried different GARCH($p$, $q$) processes with the same number of free parameters as our HARCH. The convergence of the maximum-likelihood fitting is very poor in these cases, and the final results are similar to those of GARCH(1, 1).

As in the case of HARCH, the random variable $\varepsilon_t$ of the GARCH process is normally distributed [15]. This is the resulting GARCH(1, 1) equation:

$$r_t = \sigma_t \varepsilon_t, \qquad \sigma_t^2 = 2.39 \times 10^{-8} + 0.1274 r_{t-1}^2 + 0.8446 \sigma_{t-1}^2 \tag{15}$$

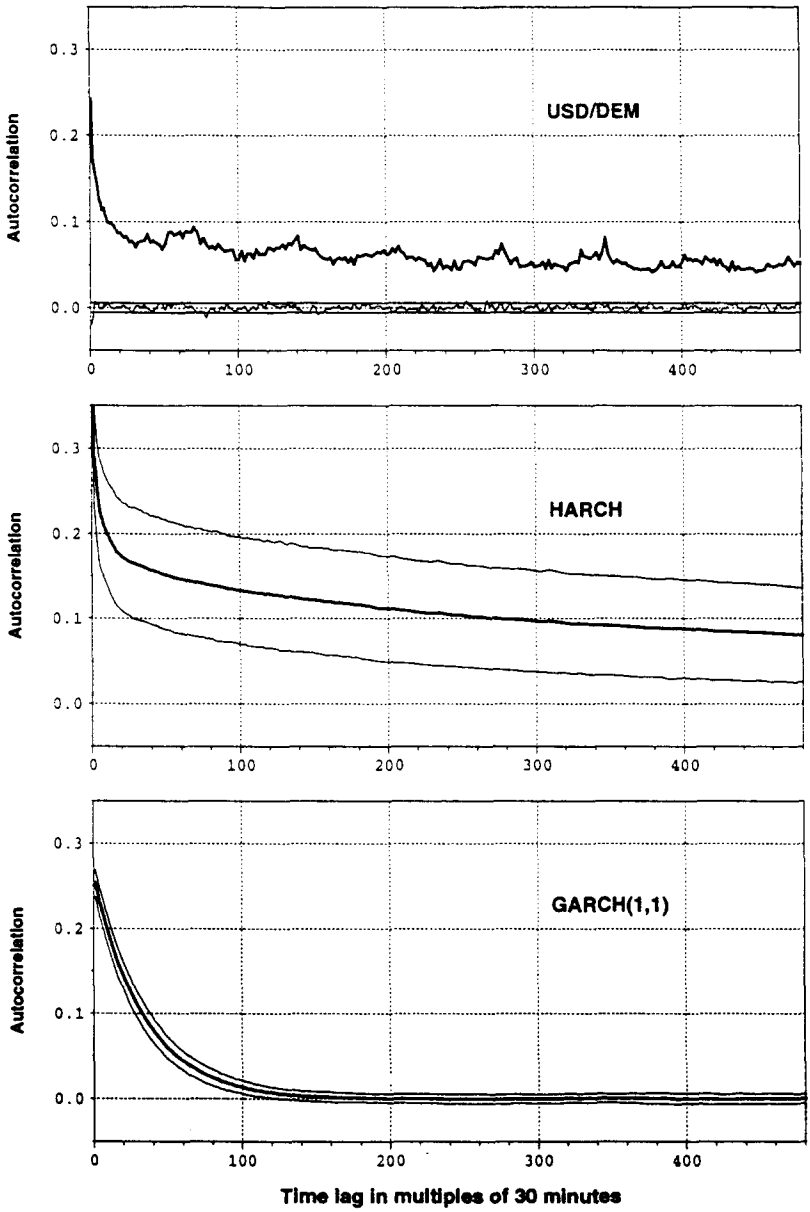
Fig. 6 shows the lagged correlation function for this GARCH series. The GARCH process is clearly unable to reproduce the asymmetry of the correlation of fine and coarse volatilities. Moreover, it is unable to model the long memory in the volatility: the correlation fades too rapidly for both positive and negative lags. The newly developed FIGARCH process (Baillie et al., 1996) can be expected to reproduce the long memory but not the asymmetry of the correlation as it is still based on time intervals of equal size.

The HARCH series is able to reproduce properties of FX series such as the long memory of volatility (Dacorogna et al., 1993; Ding et al., 1993) (see Fig. 7) as well as the fat-tailed unconditional distribution of the returns (Dacorogna et al., 1994; Loretan and Phillips, 1994) (see Fig. 8). The GARCH series has no long memory and an insufficiently fat-tailed distribution function as can be seen in both Figs. 7 and 8. The long memory effect has also been reproduced using the FIGARCH process (Baillie et al., 1996) but it is not clear that the lack of fat-tails is also corrected by such a process.

Fig. 7 shows two other interesting effects. First, the heat-wave effect (Engle et al., 1990; Dacorogna et al., 1993), the slightly higher autocorrelations at lags around integer numbers of working days, is found for USD–DEM but not for HARCH, because the HARCH variance equation does not contain terms with the selective memory of geographical markets. Second, we find a high standard deviation of autocorrelation for the 100 analyzed HARCH series. This is due to the long memory of the volatility. The clusters of volatility are so long that the number of independent clusters is relatively small, leading to high stochastic deviations. The widening of the fluctuations of the tails in Fig. 8 is also worth noticing. It is related to the fluctuations of the autocorrelation of volatility. If a big

Fig. 7. Comparative study of the autocorrelation functions. Autocorrelation functions of absolute returns over 7 years are plotted against the lag in $\vartheta$-time, for the FX rate USD–DEM and the HARCH and the GARCH processes fitted to USD–DEM. For USD–DEM, the autocorrelation function of the returns themselves is also plotted; it lies well within the confidence interval (which represents 95% confidence of the Gaussian random walk hypothesis). For each of both processes, HARCH and GARCH, 100 synthetic time series of 7 years have been generated. The bold curve represents the mean autocorrelation averaged over all 100 time series; the thin curves above and below represent the standard deviation from this mean.

[15] We also tried the estimation of GARCH with a Student-$t$ distribution function whose degree of freedom is also estimated, following Bollerslev (1987). This poses serious problems in the case of high-frequency time series: the maximum of the log-likelihood tends to be at or beyond the stationarity (IGARCH) limit.


USD/DEM
HARCH
GARCH(1,1)
Autocorrelation
0.3
0.2
0.1
0.0
0
100
200
300
400
Time lag in multiples of 30 minutes

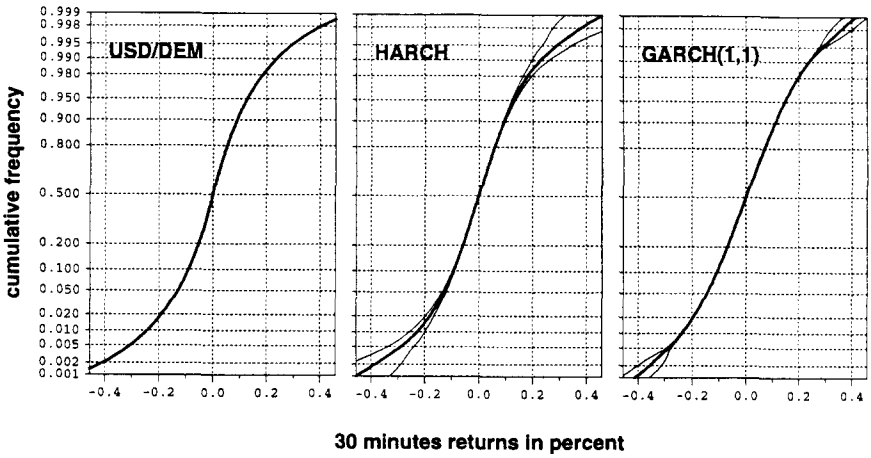



Fig. 8. Comparative study of the cumulative distribution of half-hourly returns in $\vartheta$-time. The ordinate is plotted on the scale of the cumulative Gaussian probability function. Normal distributions would have the form of a straight line on such a plot. For each of both processes, HARCH and GARCH, 100 synthetic time series of 7 years have been generated. The bold curve represents the mean of the 100 time series; the thin curves on the left and right sides represent 95% confidence of the deviation in horizontal direction.

cluster of high volatility is present in the synthetic series the tails are fatter and the autocorrelation is higher.

## 6. Conclusion

In this paper, volatilities measured with different time resolutions have been compared in a lagged correlation study. A new empirical fact in the foreign exchange market has been found: coarsely defined volatility predicts finely defined volatility significantly better than the other way around. This is true for intra-daily as well as daily time series.

From this, we conclude that the grid size of statistical volatility computation is an essential parameter which reflects the perception and the actions of different market components from short-term to long-term traders. We also conclude that the information flow between short-term and long-term traders is asymmetric: short-term traders react to coarse volatility peaks by increasing their trading activity, thus increasing the fine volatility, whereas long-term traders mostly ignore the level of fine volatility. These findings add more evidence to the hypothesis of a heterogeneous market and give an insight into the mechanisms between market components.

A new process, HARCH, has been developed to explain the asymmetric lagged

correlation as well as the long memory of volatility that has been found in earlier papers. The HARCH process differs from all other processes of the ARCH family in the size of the time intervals over which the returns of the conditional variance equation are observed. In HARCH, these time intervals vary from short-term to long-term, whereas in other ARCH-type models, these time intervals have always the same size, the size of the basic grid interval. On the other hand, all the differently sized time intervals of HARCH are 'recent' intervals ending at time $t-1$ whereas other ARCH-type models consider also some intervals in the past, ending before $t-1$.

HARCH can be formulated in such a way that the different market components are directly reflected in the terms of the variance equation. The estimated parameters of HARCH thus provide direct information about the structure of market components. All the findings about the impact of these components like the strong impact of the short-term components, the relative weakness of the 4th component (12 h), the similarity between USD–DEM and USD–CHF, or the weakness of the long-term components in GBP–USD, make economic sense and are a further confirmation of the validity of the approach.

A comparative study based on Monte Carlo simulations shows that a GARCH process is unable to reproduce the asymmetric lagged correlation, whereas HARCH reproduces the lagged correlation function of FX rates well. It also shows that Monte Carlo simulations for long memory type processes are a stringent test for random generators and produce large fluctuations in the statistical properties of the synthetical time series.

We do not consider HARCH to be the ultimate process equation. Further progress can be expected from also introducing geographical market components. Some ideas from recently proposed new ARCH variations might also be integrated in new HARCH flavors. Moreover, the study of LeBaron (1992) as well as our own recent intra-day study indicate significant trend persistence of the returns conditional to volatility. Modeling this effect is possible only with a more complicated process for the returns themselves (not only their variance). Even for such a process, the use of different interval sizes and market components as in HARCH can be recommended.

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